ability to separate words from nonwords is highly dependent on the position and completeness of token representations.

Finally, we expand the comparison of our dataset across three additional models—Llama3-8B, Mistral-7B, and Yi-6B. Our results across these models (Fig. 8) mirror the trends observed with Llama2-7B, with accuracy improving as layers deepened but peaking and gradually declining in later layers. This consistency further supports our findings that the model's ability to classify words vs. nonwords relies on nuanced patterns in internal token representations, unaffected by co-occurrence biases or morphological artifacts.

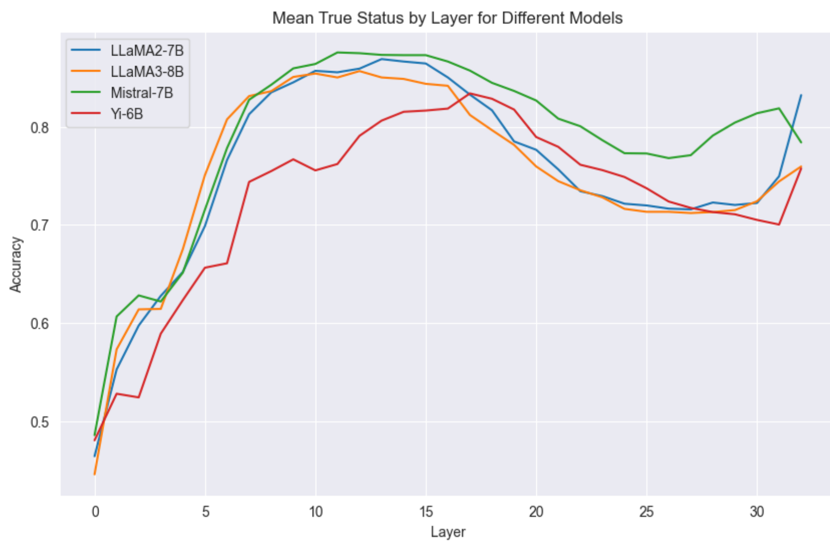

Figure 8: Mean true status accuracy by layer for distinguishing words from nonwords across multiple models (Llama2-7B, Llama3-8B, Mistral-7B, Yi-6B) using our dataset. All models exhibit similar trends: accuracy improves across initial layers, peaks in the middle layers, and declines in deeper layers. This pattern demonstrates the robustness of our findings across different model architectures and sizes.

B Statistical Analysis of Token Aggregation in Multi-token Words

We repeat the multi-token experiments Sec. 5.2 for 3- and 4-token words. Our results (Fig. 9) show a very similar trend to 2-token words (Fig. 5).

We further present the detailed statistical analysis conducted of our experiments in Sec. 5.2 to examine **token aggregation** in multi-token words compared to single-token words (control group). The objective is to investigate whether the model exhibits significantly different attention patterns between these groups, indicative of the detokenization process.

To determine significance, we perform two one-sided t-tests per layer for the Llama2-7B case: (1) Testing whether attention to prefix tokens in multi-token words is higher than to previous tokens in single-token words. (2) Testing whether attention to previous tokens in single-token words is higher than to prefix tokens in multi-token words. Table 3 shows the p-values and significance levels for each layer. Significance levels are denoted as ns (not significant), and *** ($p < 0.001$).

The results reveal that in layers 1 and 2, attention to prefix tokens in multi-token words is significantly higher than in single-token words, suggesting the early phase of **Token Aggregation**. From Layers 3 to 17, attention to single-token words is higher, indicating a shift in attention focus from prefix attention in relation regular close tokens. Notably, there are intermittent increases in attention to prefix tokens at Layers 18, 21, 25, 27, 29, and 30, possibly signaling a transfer into a prediction ensembling phase in which the attention in general is less important (Ben Artzy & Schwartz, 2024) and therefore we don't see a coherent pattern.

Layer	p-value (Multi > Single)	p-value (Single > Multi)	Significance
0	9.999×10^{-1}	5.891×10^{-5}	Single-token > Multi-token (***)
1	0.000	1.000	Multi-token > Single-token (***)
2	0.000	1.000	Multi-token > Single-token (***)
3	1.000	1.252×10^{-273}	Single-token > Multi-token (***)
4	1.000	0.000	Single-token > Multi-token (***)
5	1.000	4.466×10^{-242}	Single-token > Multi-token (***)
6	1.000	0.000	Single-token > Multi-token (***)
7	1.000	0.000	Single-token > Multi-token (***)
8	1.000	0.000	Single-token > Multi-token (***)
9	1.000	0.000	Single-token > Multi-token (***)
10	1.000	0.000	Single-token > Multi-token (***)
11	1.000	7.847×10^{-87}	Single-token > Multi-token (***)
12	1.000	6.307×10^{-303}	Single-token > Multi-token (***)
13	1.000	1.417×10^{-255}	Single-token > Multi-token (***)
14	1.000	2.397×10^{-172}	Single-token > Multi-token (***)
15	1.000	0.000	Single-token > Multi-token (***)
16	1.000	5.857×10^{-25}	Single-token > Multi-token (***)
17	1.000	1.414×10^{-152}	Single-token > Multi-token (***)
18	5.646×10^{-5}	9.999×10^{-1}	Multi-token > Single-token (***)
19	1.000	7.919×10^{-161}	Single-token > Multi-token (***)
20	1.000	2.036×10^{-286}	Single-token > Multi-token (***)
21	1.399×10^{-190}	1.000	Multi-token > Single-token (***)
22	1.000	3.647×10^{-44}	Single-token > Multi-token (***)
23	1.000	6.008×10^{-13}	Single-token > Multi-token (***)
24	1.000	1.062×10^{-15}	Single-token > Multi-token (***)
25	5.274×10^{-68}	1.000	Multi-token > Single-token (***)
26	1.000	3.670×10^{-12}	Single-token > Multi-token (***)
27	0.000	1.000	Multi-token > Single-token (***)
28	1.000	2.446×10^{-10}	Single-token > Multi-token (***)
29	1.438×10^{-202}	1.000	Multi-token > Single-token (***)
30	9.506×10^{-99}	1.000	Multi-token > Single-token (***)
31	6.731×10^{-1}	3.270×10^{-1}	Not significant (ns)

Table 3: Results of one-sided t-tests comparing attention weights between multi-token and single-token words across layers over Llama2-7B model.

(a) Llama2-7B Attention weights for 3-tokens words
(b) Llama2-7B Attention weights for 4-tokens words

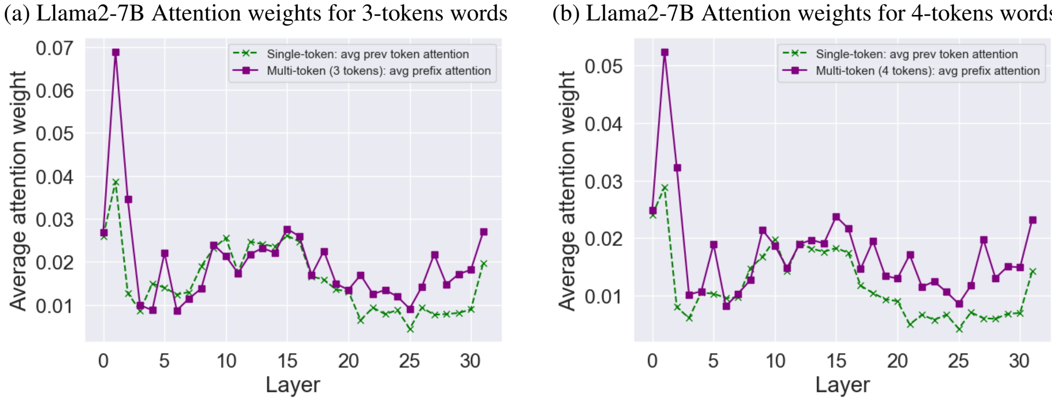

Figure 9: Analysis for 3- and 4-token words for Llama2-7B. The higher attention pattern in layers 1-2, while lower values are observed afterwards, is consistent with our results for 2-token words (Fig. 5).

In conclusion, the attention mechanism differs significantly between multi-token and single-token words, indicating that the detokenization process involves initial amplification of attention to sub-word tokens followed by a reduction as the model obtains whole-word representations.

C Word Retrieval for Single-Token and Multi-Token Words Across Models

This section presents a detailed comparison of word retrieval performance (Sec. 4) across several models (Llama2-7B, Llama3-8B, Yi-6B, and Mistral-7B) for both single-token and multi-token words. The evaluation focuses on how effectively each model retrieves the original word across layers, especially in challenging cases like artificially separated single-token words, typos, and multi-token words.

Across all experiments (Fig. 10), we observe a similar trend where the retrieval rate increases over the first several layers, peaks around the middle layers, and then decreases in the later layers. The main difference across models lies in the peak performance, especially in cases involving typos and multi-token words, where more advanced models such as Llama3-8B and Mistral-7B demonstrate superior performance in reconstructing the original word representations.

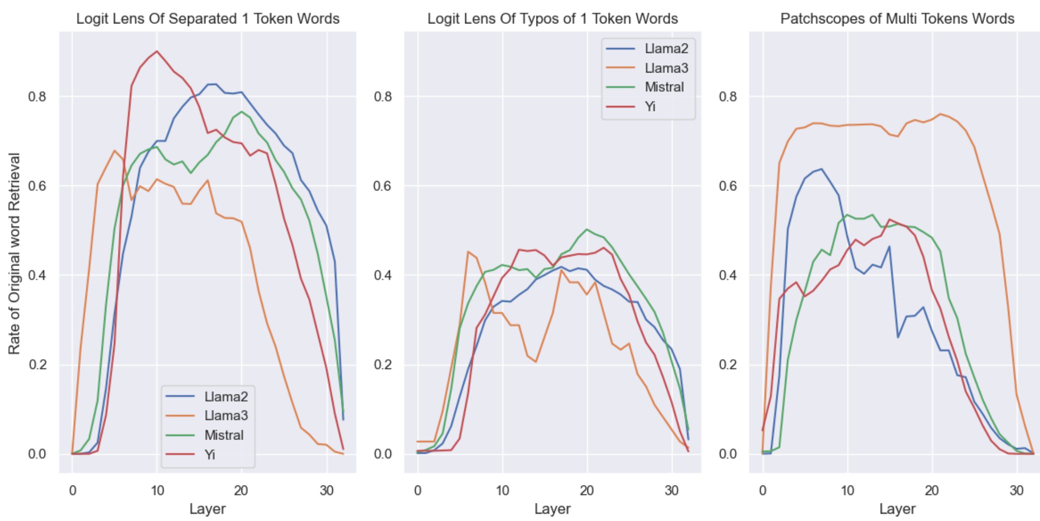

Figure 10: Layer-wise word retrieval rates for single-token and multi-token words across all models.

C.1 Cumulative Word Retrieval for Single-Token and Multi-Token Words

Fig. 11 shows the results of cumulative word retrieval (Sec. 4) across various models, focusing on both single-token and multi-token words. For each model, we analyze the ability of the LLM to retrieve the original word from sub-word tokens across its layers. The cumulative retrieval is calculated as the proportion of words that are successfully retrieved at each layer, with the percentage increasing as more words are recovered throughout the model's layers.

Single-token words The cumulative word retrieval for single-token words—those that are artificially split into sub-word tokens (via typographical errors or manual splits)—shows a rapid increase in retrieval success in the early layers. For Llama2-7B, for instance, cumulative retrieval reaches 93.2% for words split by manual intervention, and 66% for words affected by typos by the middle layers. This pattern is observed across models, with retrieval generally peaking around layers 15-20.

Multi-token words For multi-token words, which are naturally split due to being out-of-vocabulary for the tokenizer, the cumulative retrieval process follows a similar trajectory. However, in models like Llama2-7B, the retrieval peaks earlier in the model, with a cumulative retrieval rate of 77.41%. Other models like Llama3 and Yi show higher cumulative retrieval rates, suggesting improved efficiency in handling multi-token words, potentially due to larger model capacities and internal dictionaries.